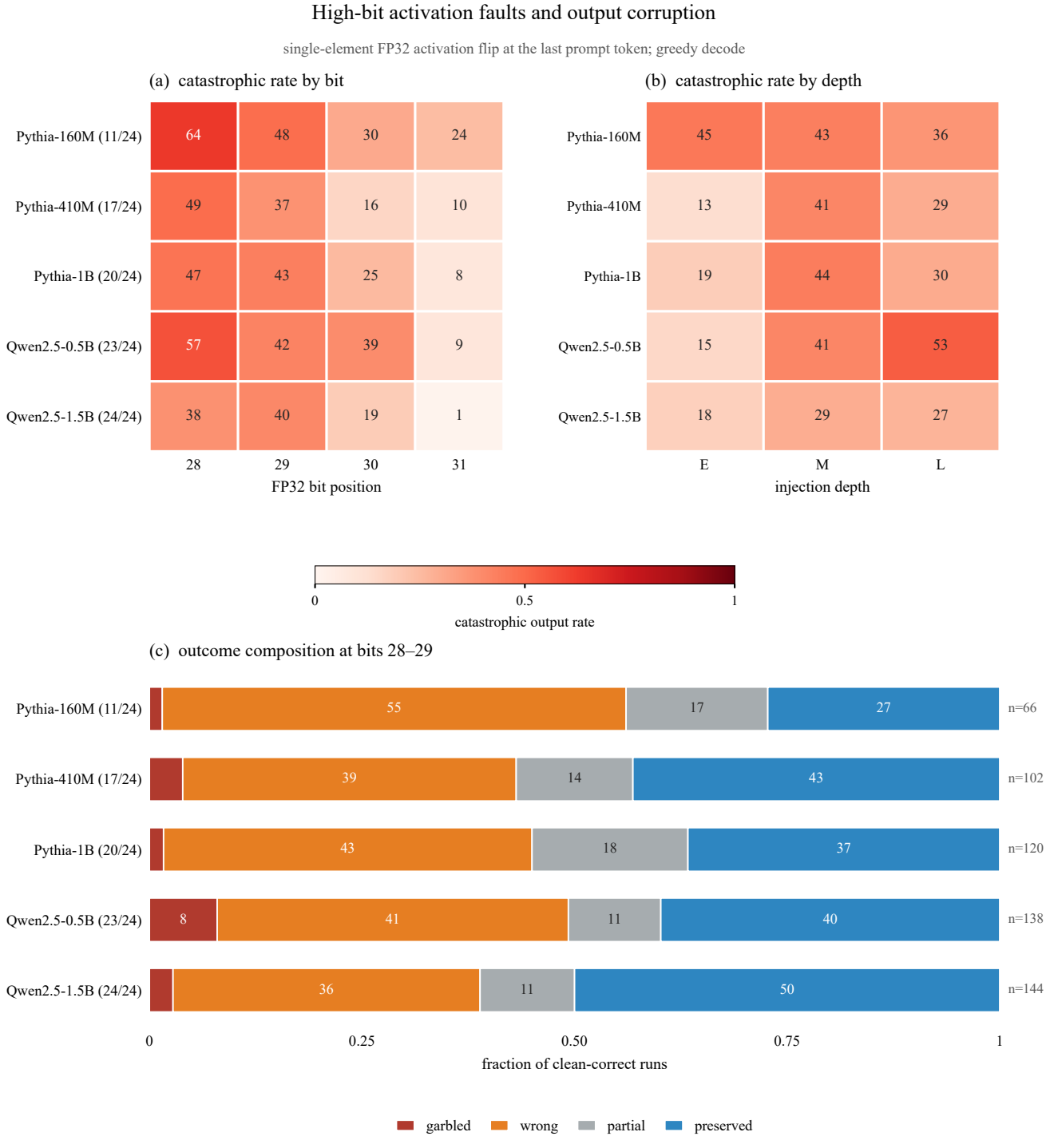
High-bit activation faults and output corruption
single-element FP32 activation flip at the last prompt token; greedy decode
(a) catastrophic rate by bit
Pythia-160M (11/24)
64
48
30
24
Pythia-410M (17/24)
49
37
16
10
Pythia-1B (20/24)
47
43
25
8
Qwen2.5-0.5B (23/24)
57
42
39
9
Qwen2.5-1.5B (24/24)
38
40
19
1
28
29
30
31
FP32 bit position
(b) catastrophic rate by depth
Pythia-160M
45
43
36
Pythia-410M
13
41
29
Pythia-1B
19
44
30
Qwen2.5-0.5B
15
41
53
Qwen2.5-1.5B
18
29
27
E
M
L
injection depth
0
0.5
1
catastrophic output rate
(c) outcome composition at bits 28–29
Pythia-160M (11/24)
55
17
27
n=66
Pythia-410M (17/24)
39
14
43
n=102
Pythia-1B (20/24)
43
18
37
n=120
Qwen2.5-0.5B (23/24)
8
41
11
40
n=138
Qwen2.5-1.5B (24/24)
36
11
50
n=144
0
0.25
0.50
0.75
1
fraction of clean-correct runs
garbled
wrong
partial
preserved
catastrophic = literal garble or answer absent; partial = answer present but altered; percentages are rounded.